

Character Array to String:-

```
public static void main(String[] args) {
    String str = " Hello Java, This is a Java Programming class";
    String str2 = "Hello all";
    String str1 = new String("Hello all");
    char arr[] = str1.toCharArray();
    for(char ch:arr){
        System.out.print(ch+" ");
    }
    System.out.println();
    //way1:
    String ans = "";
    for(char ch:arr){
        ans = ans+ch;
    }
    System.out.println(ans);
    //way2:
    String ans1 = new String(arr);
    System.out.println(ans1);
    //way3:
    String ans2 = String.valueOf(arr);
    System.out.println(ans2);
}
```

Split() → It split the string and stored in an array.

Join() → Merge the String array by given constraints.

```
public class Main {
    public static void main(String[] args) {
        String str = "Hello Java This is a Java Programming class";
        String ans[] = str.split(" ");
        for(String val:ans){
            System.out.println(val);
        }
        String ans1 = String.join("_",ans);
        System.out.println(ans1);
    }
}
```

Character Jundion:-

```
public class Main{
    public static void main(String[] args) {
        char ch1 = 'a';
        char ch2 = '1';// '0' - 48 => 49 - 48 = 1
        System.out.println(Character.isDigit(ch2));
        System.out.println(Character.isLetter(ch2));
        System.out.println(Character.isLowerCase(ch1));
        System.out.println(Character.isUpperCase(ch1));
        System.out.println(Character.toLowerCase(ch1));
        System.out.println(Character.toUpperCase(ch1));
        System.out.println((ch2-'0')+2);
    }
}
```

'0' helps to get the integer value of the character digit.

Concatenate a List of Characters

Input: arr[] = ["g", "e", "e", "k", "s", "f", "o", "r", "g", "e", "e", "k", "s"]

Output: geeksforgeeks

Explanation: combined all the characters to form a single string.

```
class Solution {
    public String charToStr(string[] arr) {
        String ans = String.Join("",arr);
        return ans;
    }
}
```

Delete Alternate Characters

Input: s = "GeeksforGeeks"

Output: "GesoGes"

Explanation: Deleted e, k, f, r, e, k at index 1, 3, 5, 7, 9, 11.

```
class Solution {
    static String delAlternate(String s) {
        String ans = "";
        for (int i = 0; i < s.length(); i += 2) {
            char ch = s.charAt(i);
            ans = ans + ch;
        }
        return ans;
    }
}
```

Count Types of Characters

Input: s = "#GeeKs01fOr@gEEks07"

Output: [5, 8, 4, 2]

Explanation: There are 5 uppercase characters, 8 lowercase characters, 4 numeric characters and 2 special characters.

```
class Sol {
    int[] count(String s) {
        int ans[] = new int[4];
        for (int i = 0; i < s.length(); i++) {
            char ch = s.charAt(i);
            if (Character.isUpperCase(ch)) {
                ans[0]++;
            }
            else if (Character.isLowerCase(ch)) {
                ans[1]++;
            }
            else if (Character.isDigit(ch)) {
                ans[2]++;
            }
            else {
                ans[3]++;
            }
        }
        return ans;
    }
}
```

Palindrome String

Input: s = "abba"

Output: true

Explanation: "abba" reads the same forwards and backwards, so it is a palindrome.

```
boolean isPalindrome(String s) {
    char arr[] = s.toCharArray();
    int i=0, j= arr.length-1;
    while(i<j){
        char temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
        i++;
        j--;
    }
    String str = String.valueOf(arr);
    return str.equals(s);
}
```

Fascinating Number

Input: n = 192

Output: true

Explanation: After multiplication with 2 and 3, and concatenating with original number, number will become 192384576 which contains all digits from 1 to 9.

$n = 192$ $a = 384$ $b = 576$

```
class Solution {
    boolean fascinating(long n) {
        long a = n*2;
        long b = n*3;
        String str = "" + n+a + b;
        if (str.length() != 9)
            return false;
        for (int i = 1; i <= 9; i++) {
            if (!str.contains(i + ""))
                return false;
        }
        return true;
    }
}
```

Extract Integers from String

Input: s = "1: Geeks for geeks, 2: geeksfor geeks, 3: forGeeksgeeks 56"

Output: [1, 2, 3, 56]

Explanation: 1, 2, 3, 56 are the integers present in s.

```

public List<String> extractInt(String s) {
    List<String> ans = new ArrayList<>();
    String str = "";
    for (int i = 0; i < s.length(); i++) {
        char ch = s.charAt(i);
        if (Character.isDigit(ch)) {
            str += ch;
            while (i + 1 < s.length() && Character.isDigit(s.charAt(i + 1))) {
                str += s.charAt(i + 1);
                i++;
            }
            ans.add(str);
            str = "";
        }
    }
    return ans;
}

```

Reverse Vowels

Input: s = "geeksforgeeks"

Output: "geeksforgeeks"

Explanation: The vowels are: e, e, o, e, e. Reverse of these is also e, e, o, e, e.

```

class Solution {
    public boolean isVowel(char ch){
        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u'){
            return true;
        }
        return false;
    }
    public String modify(String s) {
        char arr [] = s.toCharArray();
        int st = 0, end = arr.length-1;
        while(st < end){
            while(st < end && !isVowel(arr[st])){
                st++;
            }
            while(st < end && !isVowel(arr[end])){
                end--;
            }
            char temp = arr[st];
            arr[st] = arr[end];
            arr[end] = temp;
            st++;
            end--;
        }
        return String.valueOf(arr);
    }
}

```

Front-Back Transformation of String

Input: s = "Hello"

Output: Svool

Explanation: 'H' is replaced by 'S', 'e' by 'v', 'l' by 'o', and 'o' by 'l'. Therefore, the transformed string becomes "Svool".

'A' - 65

'Z' - 90

'a' - 97

'z' - 122

ch = 'b'

$=(ch - 'a') = 98 - 97$

$= 1$
 $= 122 - 1 = 121$

(char) 121

$= (char) 'z' - (ch - 'a')$

St1:- Take each character using `charAt(i)`.

St2:- Check the character is uppercase or lowercase.

St3:- If it is uppercase $\rightarrow ('Z' - (ch - 'A'))$
convert that into character.

$(char)('Z' - (ch - 'A'))$.

St4:- If it is lowercase $\rightarrow ('z' - (ch - 'a'))$
convert that into character

$(char)('z' - (ch - 'a'))$.

St5:- After the completion of loop
print the string.

```
String transformString(String s) {  
    String ans="";  
    for(int i=0;i<s.length();i++){  
        char ch = s.charAt(i);  
        if(ch>='A' && ch<='Z'){  
            ans += (char)('Z'-(ch-'A'));  
        }  
        else if(ch>='a' && ch<='z'){  
            ans += (char)('z'-(ch-'a'));  
        }else{  
            ans += ch;  
        }  
    }  
    return ans;  
}
```

HW

Capitalize First Letter of Words

Input: s = "i love programming"

Output: "I Love Programming"

1108. Defanging an IP Address

Input: address = "255.100.50.0"

Output: "255[.]100[.]50[.]0"

```
class Solution {  
    public String defangIPAddr(String address) {  
        return address.replace(".", "[.]");  
    }  
}
```

58. Length of Last Word

Input: s = "Hello World"

Output: 5

Explanation: The last word is "World" with length 5.

```
class Solution {  
    public int lengthOfLastWord(String s) {  
        s = s.trim();  
        String str[] = s.split(" ");  
        String ans = str[str.length-1];  
        return ans.length();  
    }  
}
```

709. To Lower Case

Input: s = "LOVELY"

Output: "lovely"

```
class Solution {  
    public String toLowerCase(String s) {  
        String result = s.toLowerCase();  
        return result;  
    }  
}
```

125. Valid Palindrome

Input: s = "race a car"

Output: false

Explanation: "raceacar" is not a palindrome.

```
public boolean isPalindrome(String s) {  
    int left = 0, right = s.length() - 1;  
    while (left < right) {  
        char a = s.charAt(left);  
        char b = s.charAt(right);  
        if (a >= 'A' && a <= 'Z')  
            a = Character.toLowerCase(a);  
        if (b >= 'A' && b <= 'Z')  
            b = Character.toLowerCase(b);  
        if (!((a >= 'a' && a <= 'z') || (a >= '0' && a <= '9'))){  
            left++;  
            continue;  
        }  
        if (!((b >= 'a' && b <= 'z') || (b >= '0' && b <= '9'))){  
            right--;  
            continue;  
        }  
        if (a != b) return false;  
        left++;  
        right--;  
    }  
    return true;  
}
```